

Description of Contents in Brief:

1.	Linear programming, graphical method, simplex method, two phase method and Big-M method of LPP. Degeneracy in LPP.
2.	Duality theory, dual simplex method, sensitivity analysis. Integer programming problem (IPP)- Gomory's method, Branch and Bound technique for LPP.
3.	Transportation problem (TP): Initial basic feasible solution of transportation problem, column minima method, row minima method, LCM, NWCR and Vogel's approximation methods. Optimality test for TP using UV method, unbalanced TP, maximization problem
4.	Assignment problem: Hungarian method , unbalanced problem, maximization problem
5.	Introduction of game theory, characteristics, basic definitions, Minimax-Maxmin criterion and optimality strategy, principle for mixed strategy games, graphical method and simplex method.
6	Nonlinear programming: Kuhn-Tucker conditions, Lagrange's condition and problems on Kuhn-Tucker conditions.